POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5043D | Semester V | Theory | 4 Credits

Optoelectronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043D as Optoelectronics in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Delhi’s Semiconductor Optoelectronics course and polytechnic optoelectronic curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Optical designs, laser calibrations, equipment specifications and spectrum allocations must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

> **Verified source note:** Verified sources support light-matter interaction, LEDs, Lasers (VCSEL), photodetectors (PIN, APD), and optical fibers. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Optoelectronics studies the interaction of light with electronic materials and devices for communication, sensing and power generation. It develops the ability to analyze semiconductor light sources (LEDs, Lasers), evaluate photodetectors (PIN, APD), understand optical fiber transmission and modulation, and coordinate optoelectronic system designs while respecting the technical and safety boundaries of optical and telecommunication engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Engineering physics, semiconductor devices, analog electronics, electromagnetic wave theory and basic optics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of light-matter interaction and the band structure of optoelectronic semiconductors.
2. Explain the working principles, construction and characteristics of LEDs and semiconductor laser diodes.
3. Explain the operation and performance of photodetectors, including PIN and Avalanche Photodiodes (APD).
4. Explain the propagation of light in optical fibers and the components used in optoelectronic communication systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the physics of optical processes in semiconductors and energy band engineering.	Understand
CO2	Explain the characteristics and applications of semiconductor light sources and modulators.	Understand
CO3	Explain the operation and noise performance of various semiconductor photodetectors.	Understand
CO4	Explain optical fiber transmission characteristics and optoelectronic system integration.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Light-Matter Interaction and Band Structure	Nature of light; energy bands in solids; E-k diagram; density of states; absorption, spontaneous emission and stimulated emission; direct and indirect bandgap semiconductors; heterostructures.	Approx. 14 hours	CO1	Module 1
2	Semiconductor Light Sources and Modulators	Light Emitting Diodes (LED): structure, materials, quantum efficiency, characteristics; Laser Diodes: population inversion, optical feedback, threshold current, VCSEL; Optical modulators.	Approx. 16 hours	CO2	Module 2
3	Photodetectors and Noise Performance	Photoconductors; Photodiodes: PIN photodiode, Avalanche Photodiode (APD); Responsivity and quantum efficiency; Impulse response; Noise in photodetectors (shot noise, thermal noise).	Approx. 14 hours	CO3	Module 3
4	Optical Fibers and Optoelectronic Systems	Optical fiber types (Step-index, Graded-index); Single-mode vs Multi-mode; Attenuation and dispersion; Optical couplers and connectors; Optical amplifiers (EDFA); Optoelectronic Integrated Circuits (OEIC).	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Light-Matter Interaction and Band Structure

Nature of light; energy bands in solids; E-k diagram; density of states; absorption, spontaneous emission and stimulated emission; direct and indirect bandgap semiconductors; heterostructures.

CO1 · Approx. 14 hours

Semiconductor Light Sources and Modulators

Light Emitting Diodes (LED): structure, materials, quantum efficiency, characteristics; Laser Diodes: population inversion, optical feedback, threshold current, VCSEL; Optical modulators.

CO2 · Approx. 16 hours

Photodetectors and Noise Performance

Photoconductors; Photodiodes: PIN photodiode, Avalanche Photodiode (APD); Responsivity and quantum efficiency; Impulse response; Noise in photodetectors (shot noise, thermal noise).

CO3 · Approx. 14 hours

Optical Fibers and Optoelectronic Systems

Optical fiber types (Step-index, Graded-index); Single-mode vs Multi-mode; Attenuation and dispersion; Optical couplers and connectors; Optical amplifiers (EDFA); Optoelectronic Integrated Circuits (OEIC).

CO4 · Approx. 16 hours

MODULE 1

Light-Matter Interaction and Band Structure

Nature of light; energy bands in solids; E-k diagram; density of states; absorption, spontaneous emission and stimulated emission; direct and indirect bandgap semiconductors; heterostructures.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

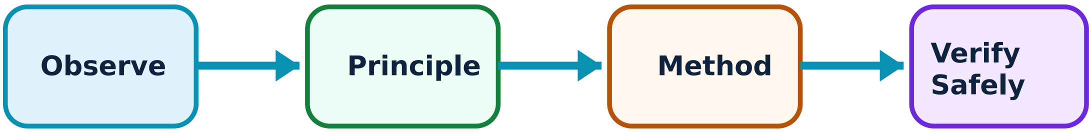
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Light-Matter Interaction and Band Structure: identify the system, state its principle, follow the correct method and verify the result safely.

1. Optical processes in semiconductors−

Light interacts with semiconductors through absorption (creating electron-hole pairs) and emission (recombination). Stimulated emission is the basis for laser action. Direct bandgap materials (like GaAs) are efficient for light emission, while indirect materials (like Si) are generally used for detectors.

Study and application method

Describe the difference between 'Spontaneous' and 'Stimulated' emission; explain why GaAs is preferred over Silicon for making LEDs.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the difference between 'Spontaneous' and 'Stimulated' emission; explain why GaAs is preferred over Silicon for making LEDs.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Optical processes are temperature-dependent. Do not operate optoelectronic devices without authorised thermal management to prevent wavelength shift or device degradation.

2. Bandgap engineering−

Bandgap engineering involves using semiconductor alloys (e.g., AlGaAs, InGaAsP) to tailor the emission wavelength and electronic properties. Heterostructures and quantum wells are used to confine carriers, improving the efficiency of LEDs and lasers.

Study and application method

Explain the concept of a 'Heterojunction' and its advantage in optoelectronics; define 'Density of States' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of a 'Heterojunction' and its advantage in optoelectronics; define 'Density of States' conceptually.

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Common mistake / precaution: Epitaxial growth of heterostructures requires precise lattice matching. Do not design multi-layer devices without authorised material compatibility and strain analysis.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Semiconductor Light Sources and Modulators

Light Emitting Diodes (LED): structure, materials, quantum efficiency, characteristics; Laser Diodes: population inversion, optical feedback, threshold current, VCSEL; Optical modulators.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

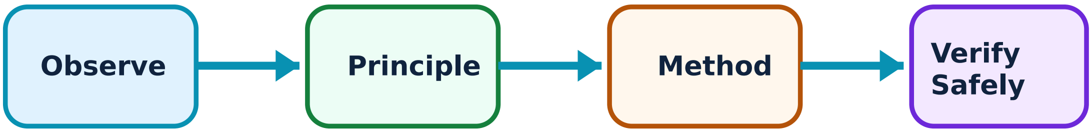
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Semiconductor Light Sources and Modulators: identify the system, state its principle, follow the correct method and verify the result safely.

1. Light Emitting Diodes (LED)−

LEDs convert electrical energy into light via injection electroluminescence. Internal and external quantum efficiencies determine overall performance. LEDs are used for indicators, displays and short-range optical communication.

Study and application method

Sketch the structure of a surface-emitting LED; list three factors that affect the 'External Quantum Efficiency' of an LED.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the structure of a surface-emitting LED; list three factors that affect the 'External Quantum Efficiency' of an LED.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-brightness LEDs generate significant heat. Do not operate power LEDs without authorised heat sinks and current-regulated drivers.

2. Semiconductor Laser Diodes−

Lasers require population inversion and optical feedback (cavity). Semiconductor lasers are compact and highly efficient. VCSELs (Vertical Cavity Surface Emitting Lasers) emit light perpendicular to the chip surface, offering advantages in array integration and high-speed modulation.

Study and application method

Explain the requirement for 'Population Inversion' in a laser; compare Edge-emitting lasers and VCSELs conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the requirement for 'Population Inversion' in a laser; compare Edge-emitting lasers and VCSELs conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Laser diodes emit high-intensity coherent light. Do not look directly into an energized laser diode without authorised eye protection and safety shielding.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Photodetectors and Noise Performance

Photoconductors; Photodiodes: PIN photodiode, Avalanche Photodiode (APD); Responsivity and quantum efficiency; Impulse response; Noise in photodetectors (shot noise, thermal noise).

Approx. 14 hours | CO3 | Understand · Apply

Learning goals

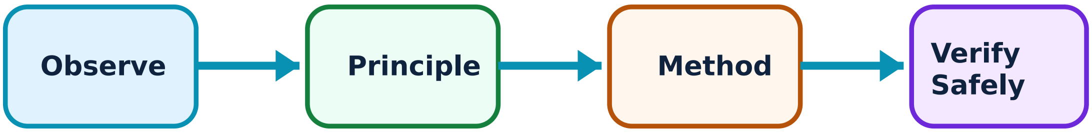
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Photodetectors and Noise Performance: identify the system, state its principle, follow the correct method and verify the result safely.

1. PIN and Avalanche Photodiodes−

PIN photodiodes have an intrinsic layer to improve absorption and speed. APDs provide internal gain through impact ionization, making them highly sensitive for long-distance communication. Responsivity measures the electrical current generated per unit of incident optical power.

Study and application method

Sketch the structure of a PIN photodiode; explain the principle of 'Impact Ionization' in an APD.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the structure of a PIN photodiode; explain the principle of 'Impact Ionization' in an APD.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: APDs require high reverse bias voltages (tens to hundreds of volts). Do not operate APDs without authorised high-voltage safety precautions and temperature compensation.

2. Detector noise and speed−

Noise limits the minimum detectable signal. Shot noise arises from the discrete nature of photons/electrons, while thermal noise comes from resistive components. The impulse response and bandwidth of the detector determine the maximum data rate of the system.

Study and application method

Define 'Responsivity' of a photodetector; list the two main types of noise that affect photodetector performance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Responsivity' of a photodetector; list the two main types of noise that affect photodetector performance.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Noise performance is critical for high-speed links. Do not design receivers without authorised noise-equivalent power (NEP) and signal-to-noise ratio (SNR) analysis.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Optical Fibers and Optoelectronic Systems

Optical fiber types (Step-index, Graded-index); Single-mode vs Multi-mode; Attenuation and dispersion; Optical couplers and connectors; Optical amplifiers (EDFA); Optoelectronic Integrated Circuits (OEIC).

Approx. 16 hours | **CO4** | **Understand · Apply**

Learning goals

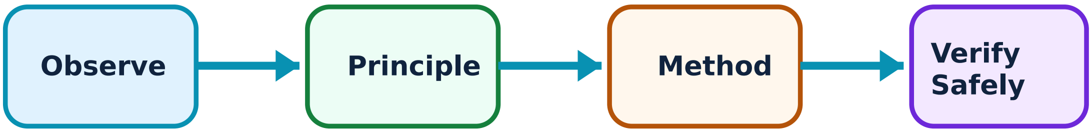
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Optical Fibers and Optoelectronic Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Fiber transmission characteristics−

Optical fibers use total internal reflection to guide light. Attenuation (loss) and dispersion (pulse spreading) limit the link distance and bandwidth. Single-mode fibers (SMF) eliminate modal dispersion, enabling long-haul, high-capacity communication.

Study and application method

Compare Step-index and Graded-index fibers; explain how 'Chromatic Dispersion' affects optical signal transmission.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare Step-index and Graded-index fibers; explain how 'Chromatic Dispersion' affects optical signal transmission.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Fiber handling requires precision. Do not bend optical fibers beyond their authorised minimum bend radius to avoid micro-bend losses or fiber breakage.

2. System integration and applications−

Optoelectronic systems integrate sources, fibers and detectors. Optical amplifiers like EDFA (Erbium-Doped Fiber Amplifier) boost signals without electronic conversion. OEICs integrate optical and electronic components on a single chip, essential for future high-speed networks.

Study and application method

Sketch the block diagram of a basic fiber optic communication system; explain the role of an 'Optical Isolator' in a laser link.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of a basic fiber optic communication system; explain the role of an 'Optical Isolator' in a laser link.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Optoelectronic systems involve complex alignment and splicing. All system installations must follow authorised optical safety and precision assembly standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Light-Matter Interaction and Band Structure

Use the concept flow to revise observation, principle, method and verification.

Module 2: Semiconductor Light Sources and Modulators

Use the concept flow to revise observation, principle, method and verification.

Module 3: Photodetectors and Noise Performance

Use the concept flow to revise observation, principle, method and verification.

Module 4: Optical Fibers and Optoelectronic Systems

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare LED and Laser diodes in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the energy band diagram of a p-n heterojunction under forward bias.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the structure of a PIN photodiode and label the P, I and N regions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the responsivity conceptually for a photodiode with a given quantum efficiency at 1550nm.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the optoelectronic components (e.g., LED, photodiode) used in a standard TV remote or an optical mouse.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Light-Matter Interaction and Band Structure

Explain Optical processes in semiconductors and state one correct method, application or precaution.−

Light interacts with semiconductors through absorption (creating electron-hole pairs) and emission (recombination). Stimulated emission is the basis for laser action. Direct bandgap materials (like GaAs) are efficient for light emission, while indirect materials (like Si) are generally used for detectors.

Answer framework: Describe the difference between 'Spontaneous' and 'Stimulated' emission; explain why GaAs is preferred over Silicon for making LEDs.

Important point: Optical processes are temperature-dependent. Do not operate optoelectronic devices without authorised thermal management to prevent wavelength shift or device degradation.

Explain Bandgap engineering and state one correct method, application or precaution.−

Bandgap engineering involves using semiconductor alloys (e.g., AlGaAs, InGaAsP) to tailor the emission wavelength and electronic properties. Heterostructures and quantum wells are used to confine carriers, improving the efficiency of LEDs and lasers.

Answer framework: Explain the concept of a 'Heterojunction' and its advantage in optoelectronics; define 'Density of States' conceptually.

Important point: Epitaxial growth of heterostructures requires precise lattice matching. Do not design multi-layer devices without authorised material compatibility and strain analysis.

Module 2 — Semiconductor Light Sources and Modulators

Explain Light Emitting Diodes (LED) and state one correct method, application or precaution.−

LEDs convert electrical energy into light via injection electroluminescence. Internal and external quantum efficiencies determine overall performance. LEDs are used for indicators, displays and short-range optical communication.

Answer framework: Sketch the structure of a surface-emitting LED; list three factors that affect the 'External Quantum Efficiency' of an LED.

Important point: High-brightness LEDs generate significant heat. Do not operate power LEDs without authorised heat sinks and current-regulated drivers.

Explain Semiconductor Laser Diodes and state one correct method, application or precaution.−

Lasers require population inversion and optical feedback (cavity). Semiconductor lasers are compact and highly efficient. VCSELs (Vertical Cavity Surface Emitting Lasers) emit light perpendicular to the chip surface, offering advantages in array integration and high-speed modulation.

Answer framework: Explain the requirement for 'Population Inversion' in a laser; compare Edge-emitting lasers and VCSELs conceptually.

Important point: Laser diodes emit high-intensity coherent light. Do not look directly into an energized laser diode without authorised eye protection and safety shielding.

Module 3 — Photodetectors and Noise Performance

Explain PIN and Avalanche Photodiodes and state one correct method, application or precaution.−

PIN photodiodes have an intrinsic layer to improve absorption and speed. APDs provide internal gain through impact ionization, making them highly sensitive for long-distance communication. Responsivity measures the electrical current generated per unit of incident optical power.

Answer framework: Sketch the structure of a PIN photodiode; explain the principle of 'Impact Ionization' in an APD.

Important point: APDs require high reverse bias voltages (tens to hundreds of volts). Do not operate APDs without authorised high-voltage safety precautions and temperature compensation.

Explain Detector noise and speed and state one correct method, application or precaution.−

Noise limits the minimum detectable signal. Shot noise arises from the discrete nature of photons/electrons, while thermal noise comes from resistive components. The impulse response and bandwidth of the detector determine the maximum data rate of the system.

Answer framework: Define 'Responsivity' of a photodetector; list the two main types of noise that affect photodetector performance.

Important point: Noise performance is critical for high-speed links. Do not design receivers without authorised noise-equivalent power (NEP) and signal-to-noise ratio (SNR) analysis.

Module 4 — Optical Fibers and Optoelectronic Systems

Explain Fiber transmission characteristics and state one correct method, application or precaution.−

Optical fibers use total internal reflection to guide light. Attenuation (loss) and dispersion (pulse spreading) limit the link distance and bandwidth. Single-mode fibers (SMF) eliminate modal dispersion, enabling long-haul, high-capacity communication.

Answer framework: Compare Step-index and Graded-index fibers; explain how 'Chromatic Dispersion' affects optical signal transmission.

Important point: Fiber handling requires precision. Do not bend optical fibers beyond their authorised minimum bend radius to avoid micro-bend losses or fiber breakage.

Explain System integration and applications and state one correct method, application or precaution.−

Optoelectronic systems integrate sources, fibers and detectors. Optical amplifiers like EDFA (Erbium-Doped Fiber Amplifier) boost signals without electronic conversion. OEICs integrate optical and electronic components on a single chip, essential for future high-speed networks.

Answer framework: Sketch the block diagram of a basic fiber optic communication system; explain the role of an 'Optical Isolator' in a laser link.

Important point: Optoelectronic systems involve complex alignment and splicing. All system installations must follow authorised optical safety and precision assembly standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Light-Matter Interaction and Band Structure.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Semiconductor Light Sources and Modulators.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Photodetectors and Noise Performance.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Optical Fibers and Optoelectronic Systems.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Nature of light; energy bands in solids; E-k diagram; density of states; absorption, spontaneous emission and stimulated emission; direct and indirect bandgap semiconductors; heterostructures..
2. Develop a complete answer or practical plan for: Light Emitting Diodes (LED): structure, materials, quantum efficiency, characteristics; Laser Diodes: population inversion, optical feedback, threshold current, VCSEL; Optical modulators..
3. Develop a complete answer or practical plan for: Photoconductors; Photodiodes: PIN photodiode, Avalanche Photodiode (APD); Responsivity and quantum efficiency; Impulse response; Noise in photodetectors (shot noise, thermal noise)..
4. Develop a complete answer or practical plan for: Optical fiber types (Step-index, Graded-index); Single-mode vs Multi-mode; Attenuation and dispersion; Optical couplers and connectors; Optical amplifiers (EDFA); Optoelectronic Integrated Circuits (OEIC)..

LAST-MILE REVISION

Quick Revision

Module 1: Light-Matter Interaction and Band Structure

Nature of light; energy bands in solids; E-k diagram; density of states; absorption, spontaneous emission and stimulated emission; direct and indirect bandgap semiconductors; heterostructures.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Semiconductor Light Sources and Modulators

Light Emitting Diodes (LED): structure, materials, quantum efficiency, characteristics; Laser Diodes: population inversion, optical feedback, threshold current, VCSEL; Optical modulators.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Photodetectors and Noise Performance

Photoconductors; Photodiodes: PIN photodiode, Avalanche Photodiode (APD); Responsivity and quantum efficiency; Impulse response; Noise in photodetectors (shot noise, thermal noise).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Optical Fibers and Optoelectronic Systems

Optical fiber types (Step-index, Graded-index); Single-mode vs Multi-mode; Attenuation and dispersion; Optical couplers and connectors; Optical amplifiers (EDFA); Optoelectronic Integrated Circuits (OEIC).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Optical processes in semiconductors	Light interacts with semiconductors through absorption (creating electron-hole pairs) and emission (recombination).
Bandgap engineering	Bandgap engineering involves using semiconductor alloys (e.
Light Emitting Diodes (LED)	LEDs convert electrical energy into light via injection electroluminescence.
Semiconductor Laser Diodes	Lasers require population inversion and optical feedback (cavity).
PIN and Avalanche Photodiodes	PIN photodiodes have an intrinsic layer to improve absorption and speed.
Detector noise and speed	Noise limits the minimum detectable signal.
Fiber transmission characteristics	Optical fibers use total internal reflection to guide light.
System integration and applications	Optoelectronic systems integrate sources, fibers and detectors.

LEARNING RESOURCES

References

Recommended optoelectronics references

1. M. R. Shenoy, Semiconductor Optoelectronics.
2. Ajoy Ghatak and K. Thyagarajan, Fiber Optics and Lasers: The Fundamentals.
3. Pallab Bhattacharya, Semiconductor Optoelectronic Devices.
4. J. Wilson and J. F. B. Hawkes, Optoelectronics: An Introduction.

Approved public optoelectronics sources

- https://nptel.ac.in/courses/115102103
- https://onlinecourses.nptel.ac.in/noc20_ph24/preview

These sources provide the verified study basis while official Course 5043D detail is unavailable; they do not replace official optical hardware designs, authorised laser calibration, certified safety standards or authorised telecom plans.